

CLC - Project Portfolio Revision

Sebastian F., Thien N., Jake N., & Christian M.

College of Science, Engineering, and Technology, Grand Canyon University

ESG-452: Capstone Project II

Prof. Russell

28-Feb-2021

CLC – Project Portfolio Revision

Project description

Harness systems designed to restrain canines in the back seat of vehicles currently on the market indicate that the seat belt systems had been crash tested. However, no crash test standard exists for testing the canine restraint. Last year, an organization of pet restraint manufacturers began working with the American Society for Testing and Standards (ASTM) to define a crash test standard that all pet harnesses would have to “reach” if companies would advertise the system as being “crash tested”. The standard minimally outlines the design of the crash canine dummy in five different sizes. Proposed Pass-Fail criteria are also documented. The deliverables for this project will include drawings for the design of the Beagle breed canine crash test dummy. The designs would include dimensions (including the center of mass), weight, material properties of the model, and how it would be assembled. Once finished with the Beagle model, the team would then create four other designs for the other sizes of canines based on the Beagle model. If time and budget permit, testing with a representative harness will be conducted according to the proposed ASTM test protocol, which is aligned with the FMVSS 213 frontal test protocol.

Design specifications and how they were defined

Most of the customer requirements used to develop the functional requirements were recorded during zoom meetings with Bruce Miller. The first requirement being that the dummy must be durable enough to withstand multiple crash tests without the quality of performance degrading. The team is unaware of exactly how many crash tests the dummy will be subjected to but was informed that sled tests will be performed at speeds up to 30 miles per hour. This information was used when selecting A1008 cold-rolled steel as the primary material for the metal skeleton of the crash test dummy. This metal is easily weldable which another requirement

of the skeletal material as welding is a big part of the process in how the team plans to assemble the bone structure of the dummy. Because the weight and size of canines varies so broadly due to the variety of different breeds, crash test dog sizes were established by the Pet Product Standards Council (PPSC). There were five different sizes established including extra small, small, medium, large, and extra large. These sizes were established by taking data from different sized canine harnesses that were already on the market and available to the public and was based on the chest girth, neck girth, and weight of the dogs that would fit each size of harness. It was made clear with the customer at an early stage that the team would only provide one crash test dummy and it would be modeled after a Beagle, which is a breed of canine that just barely falls under the medium sized dog category. The weight range for medium sized dog crash test dummies provided is anywhere from 25 pounds to 50 pounds according to the sizes established by the PPSC however since the Beagle breed is right at the border between a small and medium sized dog, a more precise weight range of 25 pounds to 30 pounds was established for the project. The chest girth of the canine dummy should be within a range of 18 inches to 30 inches and the neck girth should be between 18 inches and 23 inches. In the first proposal submitted to ASTM for canine harness crash testing standards, the center of mass was under-defined in the crash test dummy that was used. A greater defined center of mass was one of the customer requirements defined by Exponent. At first the team was asked to define the center of mass by going out and actually measuring living dogs but because none of the team members were properly equipped to do so, instead through online research, the team jointly defined the proper center of mass for a medium sized dog should be located in the upper middle section of the canine's chest cavity. Exponent also asked that the systems within the dog not be over complicated by the inclusion of instrumentation so this also became a functional requirement for the project. The proportions of

the canine crash test dummy design were largely based off of the proportions given by the American Kennel Club (AKC) for Beagles. It was determined that the overall height of the design should be between 17.72 inches and 18.9 inches and the overall length should be between 21.65 inches and 22.83 inches. These size requirements must be met to ensure that the crash test dummy properly fits in the canine harness designed for medium sized dogs. Another requirement is that the sheet metal used to form the chest cavity shape of the dummy be thin enough to be bent into the desired shape. After meeting with one of the workers from the GCU engineering shop, John Birkheimer, he informed the team that the sheet metal be no thicker than 16-gauge sheet metal. Exponent wanted the limbs of the dummy to mimic the range of motion of an actual canine so that it could be configured into a seated or laying position, so the team included both plate and ball and socket joints in the design. The team was also asked to account for the skin and fur of the canine in the dummy design^[JW2] so the team decided to include a layer of 0.25 inch thick polyurethane foam on the outside of the bone structure, and also a layer of 10 ounce duck canvas outside of the layer of foam. It is important that neither the foam or canvas layers impede the range of motion of any of the joints on the skeletal structure.

Table 1: Functional Requirements and Design Specifications

Functional Requirements	Customer Requirement	Engineering Standard	Design Specifications
Design must have a center of mass located in the middle upper chest cavity while still maintaining an overall weight of 25 lbs - 30 lbs	Ensure model weights at least 25 lbs.	ASTM test specifications.	Model can weigh up to 30 lb.s but must be at least 25 lbs.
Design must be durable enough to withstand multiple sled tests at speeds up	Use metal for the bone structure, foam for the muscle resemblance and	ASTM D - 3574 - 17	Ensure the selected foam is dense enough and flexible to wrap around the canine

to 30 mph without deforming	canvas for the skin of the model.		bond structure and provide a feel of the real thing.
No Instrumentation is to be included	Avoid instrumentation on the model.	IEEE standards are not applicable.	Do not use instrumentation on the model.
Height of design must be between 45 cm - 48 cm and a length between 55 cm - 58 cm. Design must have a chest girth between 22.5 in - 28 in	Fit appropriately on a seat belt harness	FMVSS - 209	The model must have appropriate dimensions to fit in a harness; the harness must perform to the 209 regulations.
Design must be durable enough to withstand multiple sled tests at speeds up to 30 mph without deforming	Model must withstand high force	FMVSS - 210	Model in the harness must withstand a 5000 lbs. force.
At no point during crash testing should the metal bone structure be protruding from the canvas layer representing skin/fur	Selected canvas must not tear under repeated friction between the fabric and seatbelt.	ASTM D3884 - 01	Canvas fabric should have a high durability to prevent disturbing or adding errors to the crash tests.
Range of motion should mimic that of an actual medium sized canine	Joints of dog should allow it to be configured in both seated and laying positions without the range of motion being impeded by the foam or canvas layers		Resistance of joints should be adjustable
Sheet metal should be thin enough to be bent to the desired shape and chest circumference of a medium sized dog	Chest of design should be able to properly fit a harness meant for a medium sized dog		Sheet metal used should not be thicker than 16-gauge sheet metal

Making sure the model weights at least 25 lbs. will help the team meet the ASTM test specifications. The polyurethane foam sheet is a foam that is dense and flexible enough to wrap

around the canine bone structure allowing the team to meet the ASTM D-3574-17 standards. The design of the canine crash test dummy follows the exact measurements of the beagle, therefore it will be able to fit properly on the harness; it will meet the FMVSS-209 standard. The team will be getting a harness that fits the model design in order to verify the FMVSS-210 standard. Lastly, the 60" x 36" 10 oz. Canvas Cloth will help meet the ASTM D3884-01 standard since this type of fabric is durable. However, the material will be tested to see if it actually is durable . All of these standards, however, will be tested to determine whether the designs and materials selected meet the various engineering standards.

Final design with details on how the design specifications are being satisfied by the design.

The combination of the outer shell and the mechanical skeleton of the dog will provide an accurate representation of a Beagle that can be used in K9 automotive harness testing. One functional requirement is the weight and center of mass. The design of the dog is intentionally left underweight so various weights can be inserted and removed into the chest cavity, which will affect the total weight and center of mass. The thickness of the chest cavity is also limited to 1/16th of an inch to achieve the desired shape. The properties and hardness of the chosen material at the chosen thicknesses appear to be ideal for the goal of the project. The overall goal of the design is to achieve a model that resembles the real breed of dog as much as possible without the use of instrumentation.

Timeline breakdown including task completed and new projected completion dates for remaining task

According to the current schedule, the project is 77% completed with both fall and spring semesters taken into consideration. Compared to the expected completion rate of 76.7%, the project is right on the schedule. The team has ended the research phase (100% complete),

preliminary design phase (100% complete), the final design phase (100% complete), and wrapping up the fabrication design phase (88%). Major challenges causing delay on the schedule are the pandemic restrictions and time conflicts. Time conflict makes team members double the time and effort to keep all members updated with the newest information and coming up with a plan to resolve any issues.

Within 23 weeks of development, the team has spent \$434 on metals, bolts, nuts, and joints. Other items needed to be purchased including foam and canvas for the fabrication phase, and harness, rope, safety floor tape, and other testing materials for the testing phase. The estimated cost is about \$300. In addition, the team will outsource a tailor at GCU to make the dummy's suit as the team members do not have a sufficient sewing skill. The dummy's suit will be completed within 2 weeks with a labor cost of approximately \$500. These expenses will vary slightly as team members gather more information within the next few weeks. In 8 weeks of the Fall semester, the team finalized the design drawings, purchased most of materials needed for the fabrication process, practiced welding and using the machines in the engineering shop, cut and welded separate dummy's parts. The main challenge has been the lack of hands-on experience including using the available equipment, welding and grinding. In week 9, our team will assemble the bone structure. Also, the foam and canvas will be used to fabricate the dummy's suit during week 9 and 10. From week 11 to week 12, the team will focus on testing and making modifications simultaneously. Week 13 is planned for only minor adjustments and giving the team time for any unexpected delays. Week 14 and 15 are planned for a report, presentation and engineering showcase. The snapshot below represents the most current milestones for the project. Appendix A represents the full project timeline and Appendix B indicates the timeline breakdown structure.

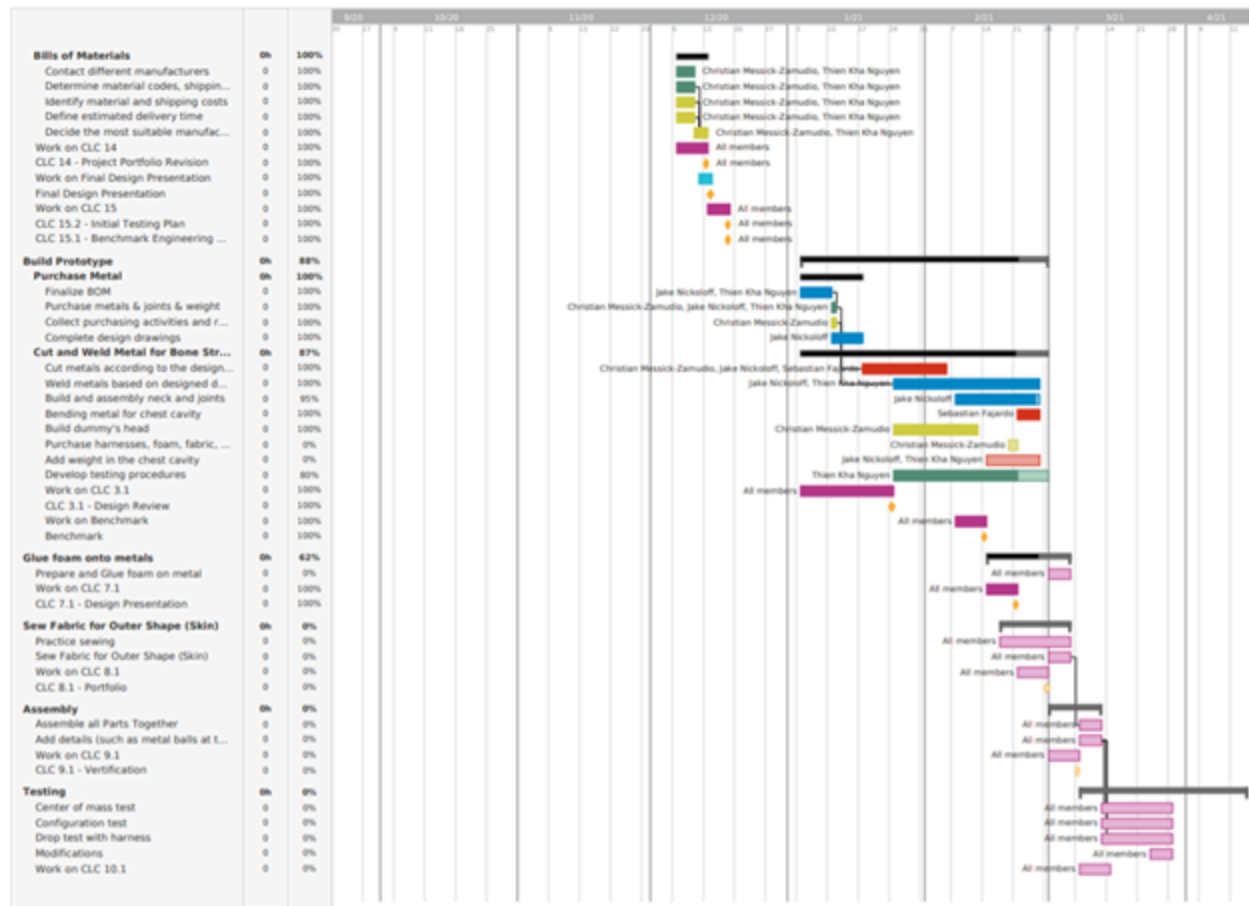


Figure 1: Project Timeline Snapshot

Plan for obtaining and using resources as well as cost estimates

DATE OF PURCHASE	ORDER #	QUANTITY	MATERIAL PURCHASED	PRICE	WEBISTE
11/18/2020	167816	1	Sample Request	\$5.95	https://www.bigduckcanvas.com/
12/2/2020	6766857	1	12-piece foam material sample pack	\$47.47	https://www.mcmaster.com/foam
12/2/2020	6766857	1	6-piece foam hardness sample pack	\$32.97	https://www.mcmaster.com/foam
12/2/2020	6766857	1	Shipping and tax	\$15.41	https://www.mcmaster.com/foam
01/13/2021	4920615	1	CR Steel Sheet 16 GA (36" x 60")	\$73.63	Industriual Metal Supply
01/13/2021	4920615	1	1018 CR Steel Rnd 5/8 x10FT ASTM A 108 (60")	\$18.37	Industriual Metal Supply
01/13/2021	4920615	1	Steel Tube Sq 1 1/2 X 1 1/2 x .250 ASTM A 500 (60")	\$62.20	Industriual Metal Supply
01/13/2021	4920615	1	1018 CR Steel Fit Bar R/L 5/8 X 3 ASTM A 108 (60")	\$96.88	Industriual Metal Supply
01/13/2021	4920615	1	Shipping and tax	\$21.59	Industriual Metal Supply
01/15/2021	485	1	Flat Washer Zinc 3/8 25PK	\$3.98	Home Depot
01/15/2021	485	1	Hex Nut Zinc 3/8 x 25PK	\$3.56	Home Depot
01/15/2021	485	1	Hex Bolt Zinc 3/8 X1	\$2.40	Home Depot
01/15/2021	485	1	1/2 Drive Universal Joint	\$21.94	Home Depot
01/15/2021	485	1	Shipping and tax	\$2.74	Home Depot
02/04/2021		1	Spirit Plush Stffued Animal	\$14.97	Wal-mart
02/04/2021		1	Spam Ripper	\$1.97	Wal-mart
02/04/2021		1	Tax	\$1.97	Wal-mart
02/04/2021		1	Sample Duck Cloth	\$5.97	Wal-mart
02/05/2021	5935676	2	Hollow Steel Ball	\$27.02	Kingmetals.com
02/26/2021	114-2916778-3236247	1	Testing Harness	\$15.99	Amazon
02/26/2021	114-2916778-3236247	1	Yellow Caution Tape	\$12.98	Amazon
02/26/2021	114-2916778-3236247	1	10 oz Duck Canvas Cloth	\$54.95	Amazon
02/26/2021	114-2916778-3236247	1	Polyurethane Foam Sheet	\$19.98	Amazon
02/26/2021	114-2916778-3236247	1	Ball Socket Joints	\$73.20	Amazon
02/26/2021	114-2916778-3236247	1	Testing Rope	\$12.99	Amazon
02/26/2021	114-2916778-3236247	1	Tax	\$16.37	Amazon

Figure 21: Bill of Materials

All of the needed materials have been purchased and are shown in the bill of materials.

The rope, yellow caution tape, and harness are all to be used during the projects testing phase.

The rest of the materials have been or are going to be used to construct the metal, foam, and fabric layers of the crash test dummy.

Complete risk assessment

Most of the risk about the project would mainly affect the timeline that has been set for the completion of the project. Therefore, a contingency plan needs to be implemented in case the deadlines are not met on time. However, as seen in Table 1, the likelihood of these risks occurring is as high as 3, meaning that the chances of it occurring are not too high.

Table 1: Previous Risk Assessment

#	Prospective Risk	Who is Affected by the risk	Likelihood of Occurrence	The severity of the Risk	Defensive Measures
1	Unable to meet the specifications of the ASTM test if the model weighs less than 25 lbs.	Capstone group and Exponent's.	3	Extreme	Ensure that the canine crash test model does not go under 25 lbs. Make sure to stay with the same material for the bone structure or one with similar properties such as weight.
2	Fabricating polyurethane exteriores can exceed the budget and scope of the project.	Capstone group and Exponent's.	1	Medium	Identify another foam that has similar properties to polyurethane but it's more within the budget's range.
3	Canvas creating lots of frictional contact with the harness	Capstone group and Exponent's.	2	Low	Place a rolling element on the surface of the harness that has contact with the dummy or perhaps use a type of lubricant that would not alter the results of the analysis.
4	Decreasing the weight of the model.	Capstone group and Exponent's.	3	Extreme	Make sure the material used for the bone structure does not modify the weight and center of mass of the canine crash test model.

5	Bone structure being damaged and facing protrusions.	Capstone group and Exponent's.	3	High	Enforce the areas with highest risks of protrusions and cover them well with the appropriate canvas and foam.
---	--	--------------------------------	---	------	---

Table 1 displays the previous risks the team was facing during the previous portfolio revision. However, those risks do not apply to what the team is currently working on since they have already been avoided or they are not applicable anymore.

Table 2: Updated Risk Assessment

#	Prospective Risk	Who is Affected by the risk	Likelihood of Occurrence	The severity of the Risk	Defensive Measures
1	Unable to follow the proposed timeline and complete all tasks on time.	Capstone group and Exponent.	3	High	Ensure the team meets the goals for each week. If something comes up, team members should communicate to prevent any setbacks.
2	Unable to reserve the high speed camera on time. Otherwise, it would take longer to do one of the verification tests	Capstone group.	2	High	Ensure one of the team members reserves the high-speed camera through the appropriate channels.
3	Inability to finish the dummy assembly would delay the canva suit for two to three weeks.	Capstone group and Exponent.	2	Extreme	Ensure all team members, in charge of manufacturing, finish off the bone structure assembly of the dummy.

4	Inability to review the verifications test would affect the preparations for the verification test presentation and delay the project.	Capstone group	1	High	Meet up with the capstone mentor and class instructor on time and review the various verifications tests.
5	Crash test dummy completely breaking from a drop test	Capstone group and Exponent	3	Extreme	Ensure that the dummy model is well-protected and attach the harness properly.

The following ratings are used to measure the severity of the risk: Minimal, low, medium, high, and extreme, The likelihood of the occurrence is measured from 1 to 5. Where one is very unlikely for it to occur and five very likely for it to occur.

Clearly documented revision history

Design 1

The first design was a simple sketch of a Labrador. The goal of this was to brainstorm different joints and points of mass required to resemble the size and shape of a common dog breed.

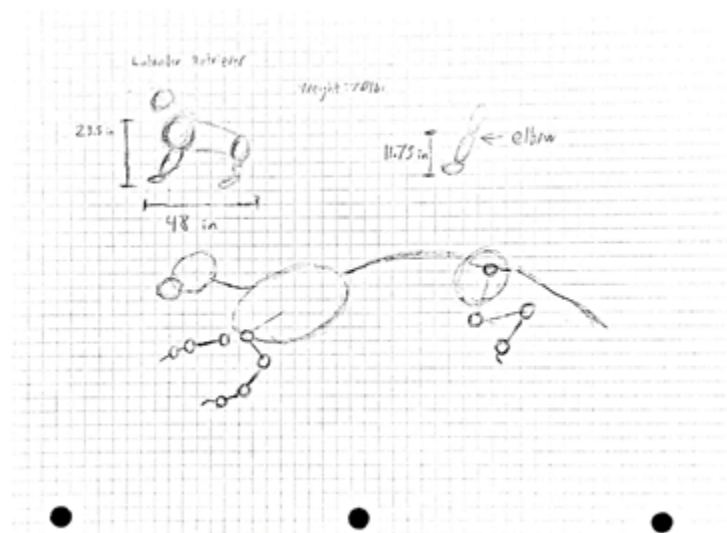


Figure 2. First Design Concept Sketch

Design 2

This design was a quick put together SolidWorks model to figure out the best methods of making each of the parts. No dimensions or weighting was considered in this model.

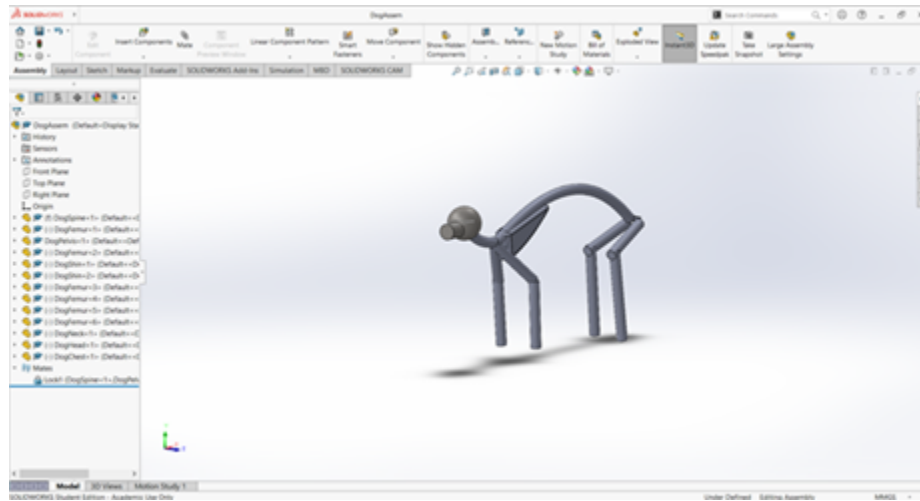


Figure 3. First Design Concept SolidWorks Model

Design 3

This model shows a complete redo of the design from the previous submission. The model now featured working plate joints on the legs and ball and socket joints on the hips and shoulders. The size of the ribcage was also increased noticeably to a more accurate size of a Beagle, which is the final choice for what the dog would be modeled after. The shoulder blades are also a new addition that allowed proper range of motion of the front legs.

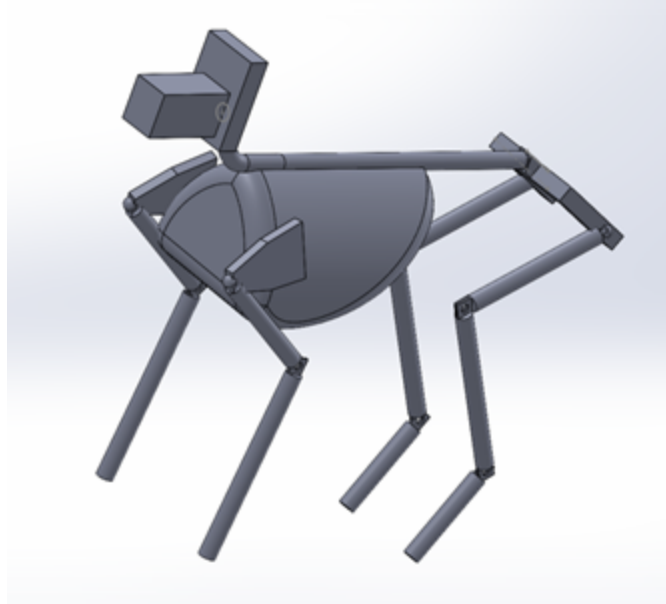


Figure 4. Current Design Concept - Crash Test Dummy Prototype version 2

Design 4

This represents how the design was able to achieve a sitting position with the addition of the joints mentioned previously. The head position was also fixed by straightening out the neck. There is also now a drawing that shows the model is within the range of dimensions for the common beagle.



Figure 5. Current Design Concept - Dog Prototype 2 (Sitting)

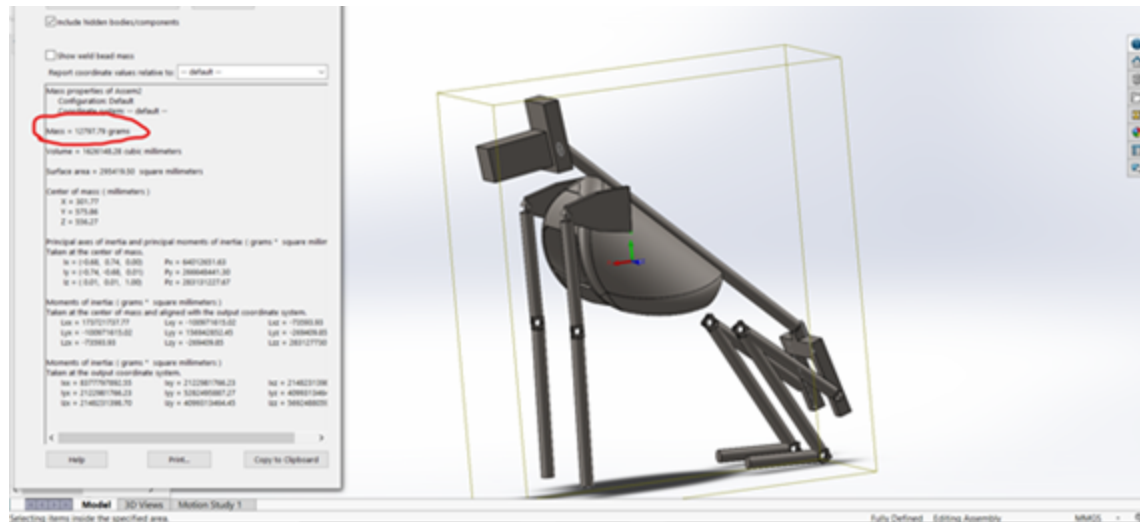


Figure 7. Current Design Concept - Dog prototype 2.1

Design 6

The model now includes a modified neck assembly that would be capable of deflection of a real beagle's neck. It is an integrated design that uses real crash test dummy necks. The neck has also been hollowed out for center of mass and weight reasons.



Figure 8. Current Design Concept - Dog prototype 2.2

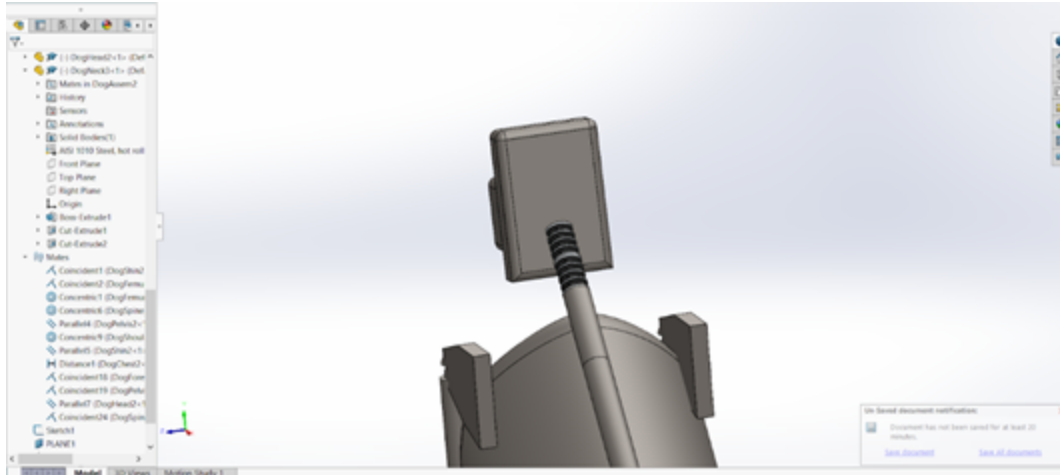


Figure 9. Current Design Concept - Dog prototype 2.2 neck

Design 7

This design goal was to select possible joint models to connect the bone structure together. The joints are crucial to the range of motion. The ball and socket model and knuckle joint were two possible options selected to model.



Figure 10. Current Design Concept - Ball and Socket Joint

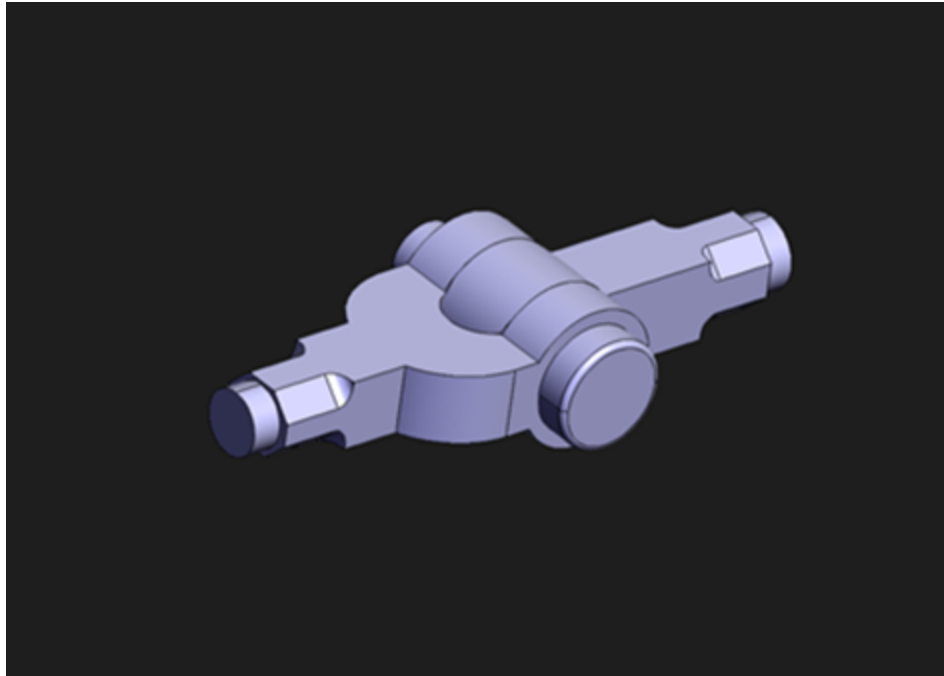


Figure 11. Current Design Concept - Plate/Knuckle Joint

Design 8

This design includes a modification to the shoulder position/design, as well as actual models for the ball and socket and plate joints. This orientation will simulate the dog's range of motion as accurate as possible. The socket also includes a possible screw that will be drilled into the outside to allow adjustable friction. The plate joints are a screw, bolt, and two washers. The shoulder is also reinforced with a pipe that will be attached to the spine. The mass of the dog was 28.2 lbs.



Figure 12. Current Design Concept - Dog prototype version 3.0

Design 9

This is the final design submitted last semester. The chest has been swapped out for a 16 gauge sheet metal thickness for a better representation of what is more realistic to accomplish in real-life. This however decreased the weight of the model from 28 to 14.8 lbs. This can be solved through placements of extra weight inside the chest. There is also a drawing present that shows the model is within the desired range of dimensions for the common Beagle.



Figure 13. Current Design Concept - Dog prototype Version 3.1

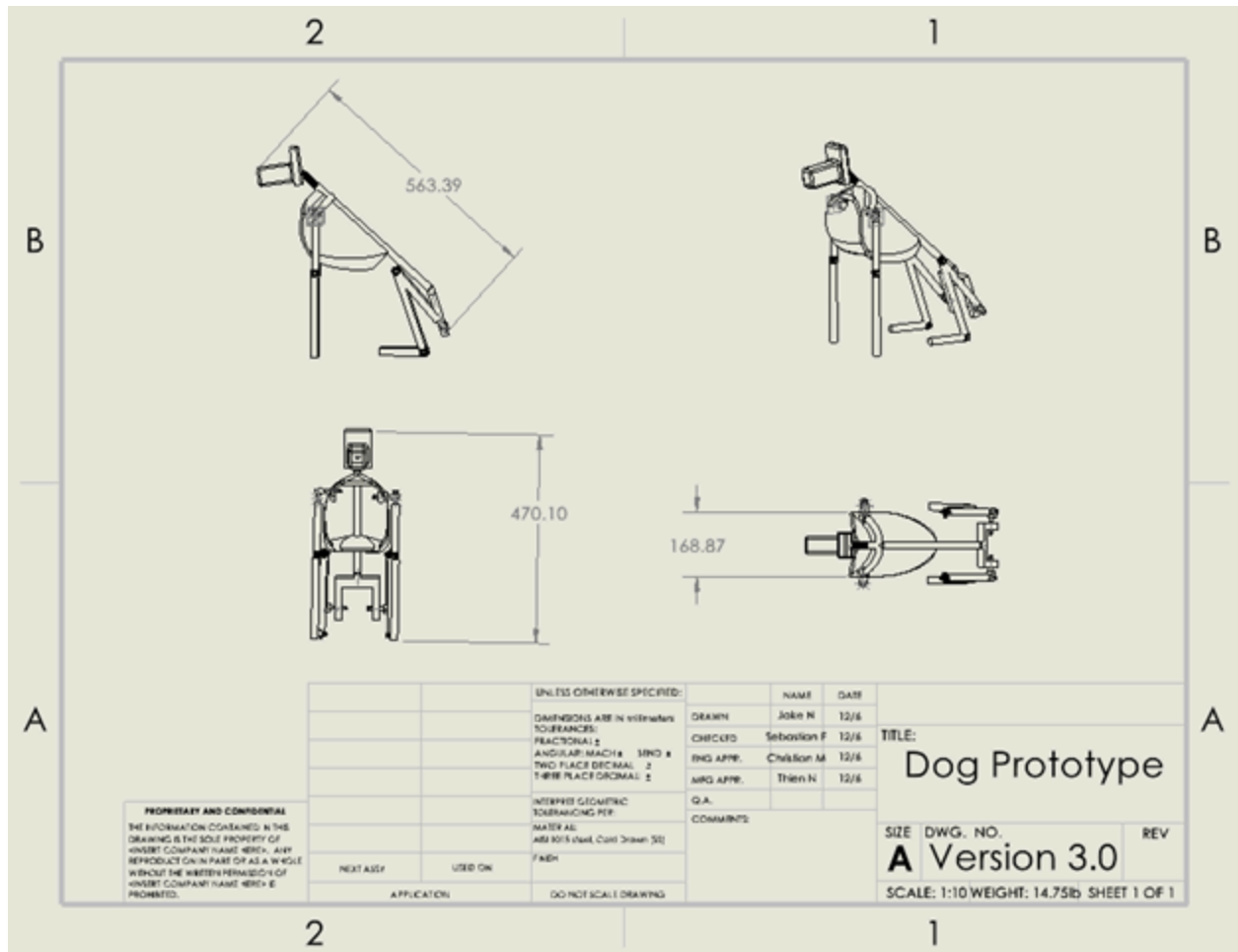


Figure 14. Current Design Concept - Dog prototype Version 3.1 - Drawing

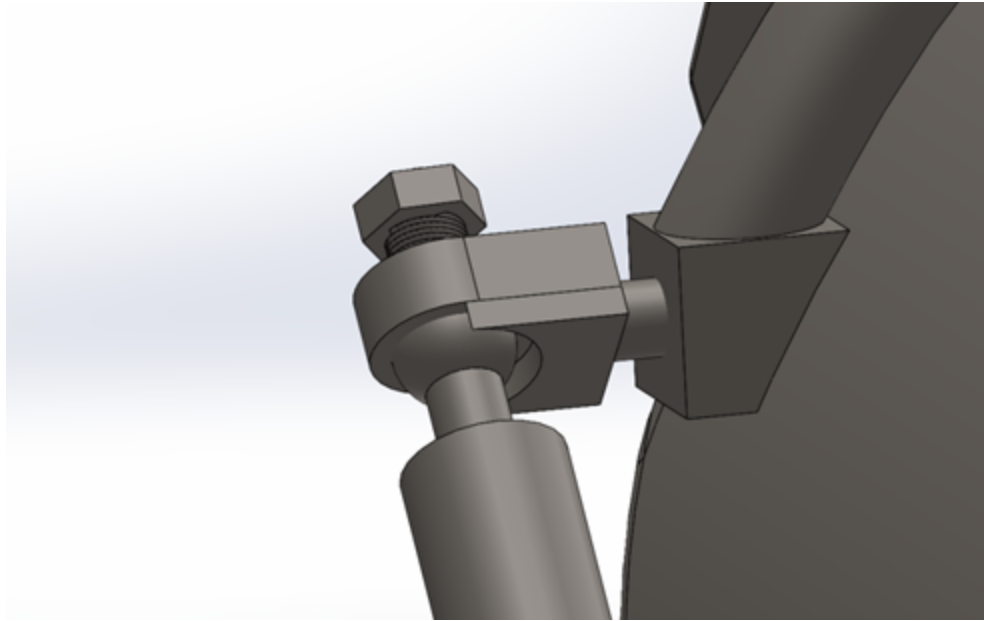


Figure 15. Current Design Concept - Dog prototype Version 3.1 – Ball & Socket Joint

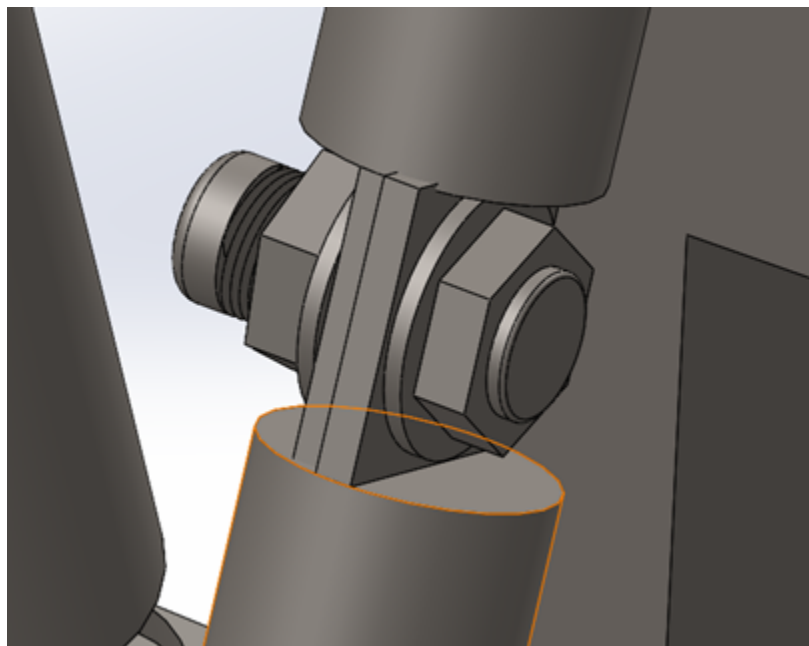


Figure 16. Current Design Concept - Dog prototype Version 3.1 – Plate Joint

Design 10

The final design of the crash test dummy dog is pictured below. One final adjustment was made by adding two front paws to the front legs of the dog. This will prevent the sharp corners from puncturing the seat of the vehicle being tested. These balls will be made of steel and welded to the end.



Figure 17. Current Design Concept - Dog prototype Version 3.2

Complete Design

The drawing of the complete design features a table of all components as well as the necessary quantity for each. The parts are also labeled on the dog and are easily identifiable by number.

The material is also stated, and the document is initialed and dated by the Design Lead Jake Nickoloff.

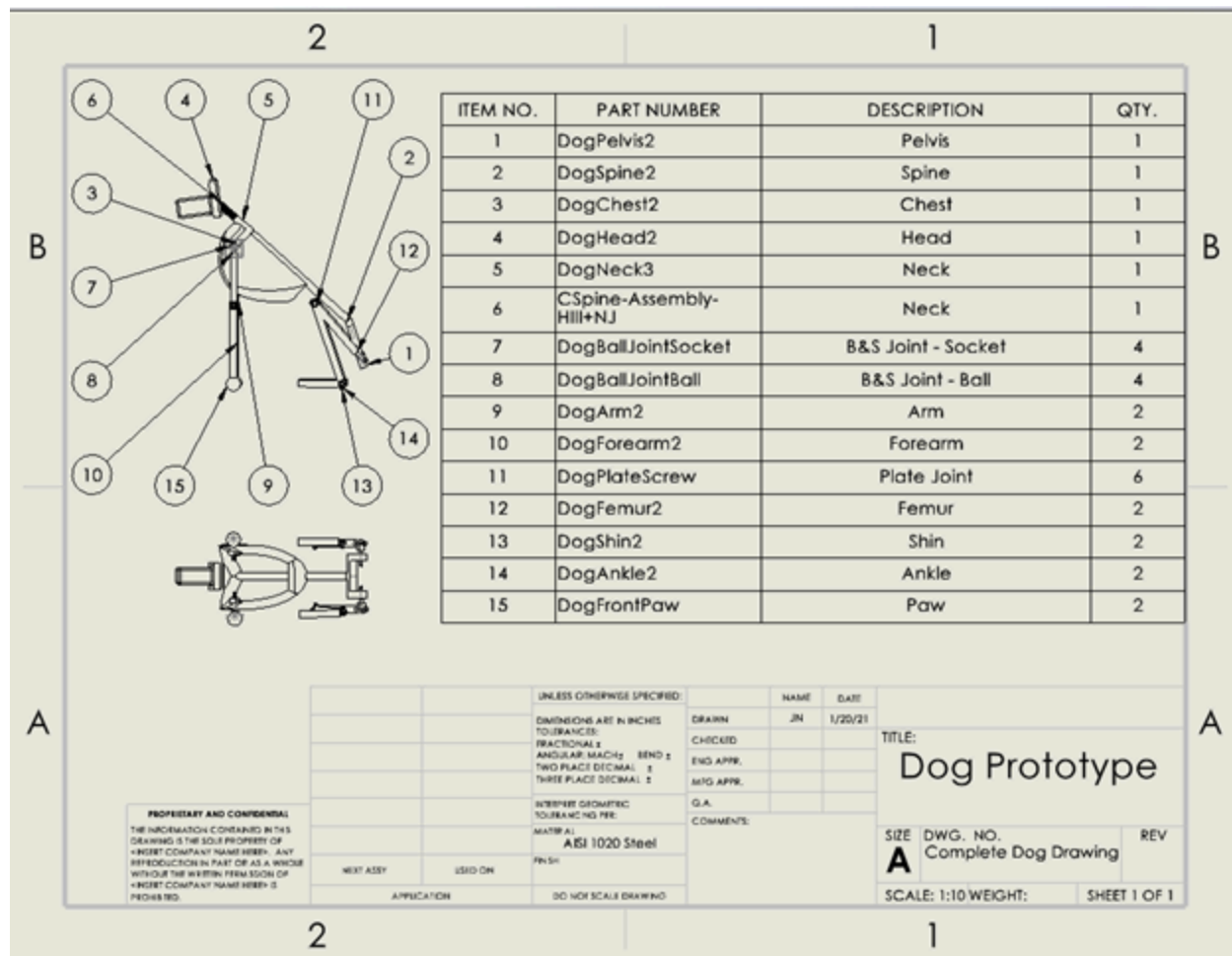


Figure 18. Dog Prototype – Complete Dog Drawing

Prototype Progress

The outer shell of the metal skeleton of the dog will consist of canvas fabric skin and foam to act as muscle filler. The team has been continuing to work in the GCU engineering shop to complete the first prototype. Since there are no software or electrical components to the design, the metalworking aspects of the shop such as MIG welder, sheet metal bending, drill press, and the cold cut saw have been used to construct the inner skeleton of the dog. The team has completed all bone segments with corresponding plate joints, chest cavity, face, spine, neck, and pelvis.



Figure 19. Bone Segments with Plate Joints

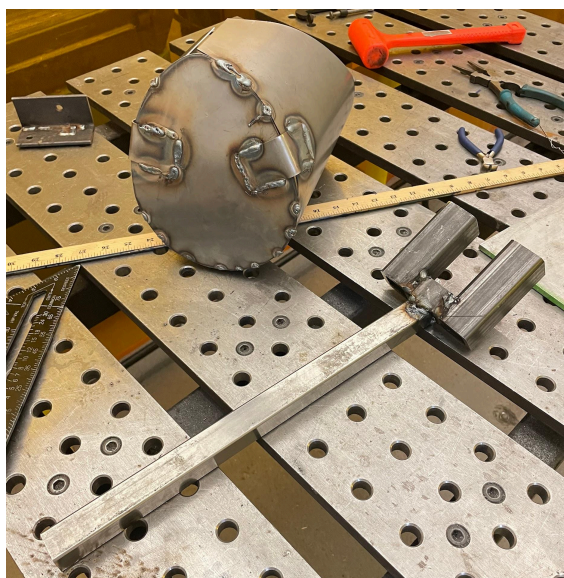


Figure 20. Completed Chest Cavity and Spine/Pelvis Combination

Appendix

Timeline Breakdown

Milestone	Planned	Forecasted	Actual	Status
Determined what aspect of canine bone structure to include in design	10/12/20	10/18/20	10/16/20	completed
Determined how lower legs, upper legs, and torso are to be connected	10/12/20	10/18/20	10/16/20	completed
Determined the requirements and time for IACUC Animal Study Approval	10/12/20	10/14/20	10/14/20	completed
Classified dog breeds into different categories based on weight, harness size and chest girth	10/12/20	10/16/20	10/16/20	completed
Determine types of joints and neck alternatives for the dummy	10/12/20	11/15/20	12/07/20	completed
Research material alternatives for dummy's skin and bone	10/12/20	11/15/20	12/02/20	completed
Identify weight distribution and center of mass	10/12/20	11/15/20	12/02/20	completed
Design outer shape for the dummy	10/12/20	11/15/20	12/02/20	completed
Design bone structure for the dummy	10/12/20	11/15/20	11/30/20	completed

Canine dummy assembly	10/30/20	12/13/20	12/13/20	completed
Identify potential manufacturers	10/30/20	12/13/20	12/13/20	completed
Develop bills of materials	10/30/20	12/13/20	12/13/20	completed
Prepare for final design presentation	10/30/20	12/13/20	12/13/20	completed
Purchase metals, joints and weights	01/04/21	01/11/21	01/15/21	completed
Plan for initial testing	01/04/21	02/26/21		behind schedule
Build the bone structure	01/04/21	02/26/21		on schedule
Add weights to the chest cavity	01/04/21	02/26/21		on schedule
Make adjustments to the joints	01/04/21	02/14/21	2/11/2021	completed
Weld the joints onto bone structure	01/04/21	02/26/21		behind schedule
Purchase harnesses, foam and fabric	01/04/21	02/21/21	02/26/21	completed
Perform first tests	01/04/21	03/08/21		on schedule
Deliverable	Planned	Forecasted	Actual	Status

1st Solidworks design	10/12/20	10/18/20	10/16/20	completed
Updated bill of materials	10/12/20	10/18/20	10/18/20	completed
Dog breed classifications and selections based on weight, chest girth and harness size	10/12/20	10/18/20	10/16/20	completed
Project portfolio	10/12/20	10/18/20	10/18/20	completed
Design Presentation	10/18/20	10/21/20	10/21/20	completed
Joint mechanisms to be used in neck, torso, and legs	10/12/20	11/15/20	12/02/20	completed
Outer shape design	10/12/20	11/15/20	12/02/20	completed
2 nd bone structure design	10/12/20	11/15/20	12/02/20	completed
Canine dummy assembly	10/30/20	12/13/20	12/13/20	completed
Bills of materials	10/30/20	12/13/20	12/13/20	completed
Final design presentation	10/30/20	12/14/20	12/14/20	completed
Initial testing plan	01/04/21	02/26/21		behind schedule
Bone structure prototype	01/04/21	02/26/21		on schedule

First testing results and analysis	01/04/21	03/08/21		on schedule
------------------------------------	----------	----------	--	----------------